



Construction Bidding & Procurement Marketplace

A multi-tenant, two-sided marketplace that takes construction procurement online end to end — companies post projects built from reusable scope packages, contractors win work through paid, priced bids, and awarded contracts run in a shared workspace with payments, reviews, disputes, and real-time collaboration.

INDUSTRY	SCOPE	MODEL	PLATFORM
Construction · Procurement	Full-stack platform	Multi-tenant SaaS	AWS · NestJS · React

Confidentiality. Sanitized portfolio case study. The client identity, product name, brand assets, proprietary business rules, and internal source are withheld under NDA. Visuals are original architecture diagrams; no client screenshots, data, or credentials are reproduced. Endpoint/module counts are drawn from the delivered codebase; commercial figures are intentionally omitted.

© 2026 Rohail K. Malhi — All rights reserved. Proprietary work; **not open-source**. The frameworks, architecture, and artifacts shown here may not be copied, adapted, or repurposed. Shared solely to evaluate the author's experience.

AT A GLANCE

CHALLENGE

Construction procurement runs on email, phone, and spreadsheets — no trusted marketplace, unstructured bidding, no commitment on either side, work scattered after the handshake, and no reputation layer.

SOLUTION

A multi-tenant, two-sided marketplace: post projects from reusable scope packages, win work through pay-to-bid, bind contracts through pay-to-award, then deliver in a shared workspace — with reviews and per-org RBAC throughout.

RESULT

A production, cloud-native platform — ~25 modules, ~190 endpoints — with money and trust built into the transaction, multi-company tenancy from day one, and an installable, real-time client.

Two-sided

Discover → bid → award
→ deliver, one platform

Pay-to-bid + award

Stripe-gated
transactions with auto-
refunds

~190 / ~25

REST endpoints across
NestJS domain modules

Multi- tenant

Six-role RBAC + custom
permission groups

CHALLENGE

Procurement stuck offline

Sourcing subcontractors — or finding construction work — still runs on informal, disconnected tools. That informality creates friction on both sides of every deal:

- **No trusted marketplace.** Owners and contractors have no single, searchable place to find each other, ranked by location, reputation, and track record.
- **Unstructured bidding.** Scope lives in scattered documents; proposals arrive as emails and PDFs with no consistent, comparable, priced format.
- **No commitment.** Free bidding invites low-quality spam; a verbal "you're hired" isn't a binding award. Nothing signals that a bid — or an award — is serious.
- **Work scatters after the handshake.** Tasks, budgets, material requests, files, and conversations spread across half a dozen tools.
- **No reputation layer.** Without verified reviews, it's hard to tell a reliable contractor from an unproven one.
- **Multi-company reality ignored.** Professionals work across several companies and need different permissions in each — consumer tools don't model this.

SOLUTION

A monetized, multi-tenant marketplace

A cloud-native platform — an installable React SPA over a modular NestJS API — carrying the full commercial lifecycle, with money and trust built into the transaction rather than bolted on.

- **Post & discover.** Companies assemble projects from reusable *tender* scope packages via a guided wizard and publish to a searchable marketplace ranked by proximity, rating, completed projects, and profile strength.
- **Pay-to-bid.** Contractors submit structured, priced bids; an upfront Stripe payment keeps the funnel serious — a bid isn't live until payment clears (confirmed by webhook).
- **Pay-to-award.** The hiring company pays an acceptance fee to award; on confirmation the bid becomes a binding active contract and the project goes live. Rejecting a paid bid triggers an automatic full refund.
- **Post-award workspace.** Tasks & kanban, three finance ledgers (payments, cash flow, cost control), inventory & material requests, reviews, disputes/claims, and real-time messaging.
- **Multi-tenant RBAC.** Header-driven organization context and a layered guard stack (account type → role → fine-grained permission) let one identity operate across many companies, each with its own role matrix.

Architecture at a glance

```
React 18 SPA (Vite · TypeScript · RTK Query, org-scoped API client)
  | Bearer JWT + x-organization-id header on every request
Request pipeline: JWT middleware → Org-context middleware → guard stack (account-type · role · permission)
NestJS 10 API (~25 modules · ~190 endpoints): Marketplace · Contracts · Orgs/Members · Project Ops · Messaging
  | Data: MongoDB Atlas (Mongoose)      | Media: AWS S3 + sharp → WebP
Integrations: Stripe (pay-to-bid / pay-to-award / refunds) · Firebase auth + 2FA · PubNub chat · email · scheduler
Delivery: AWS App Runner · Docs: Swagger / OpenAPI
```

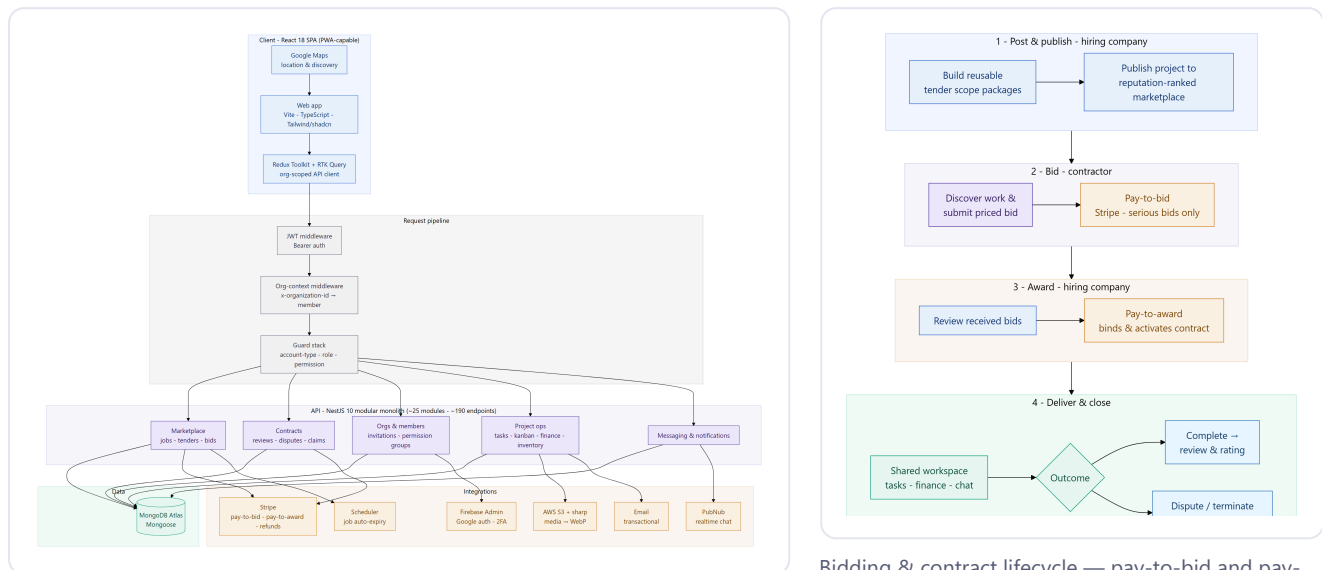
Technology



THE PLATFORM

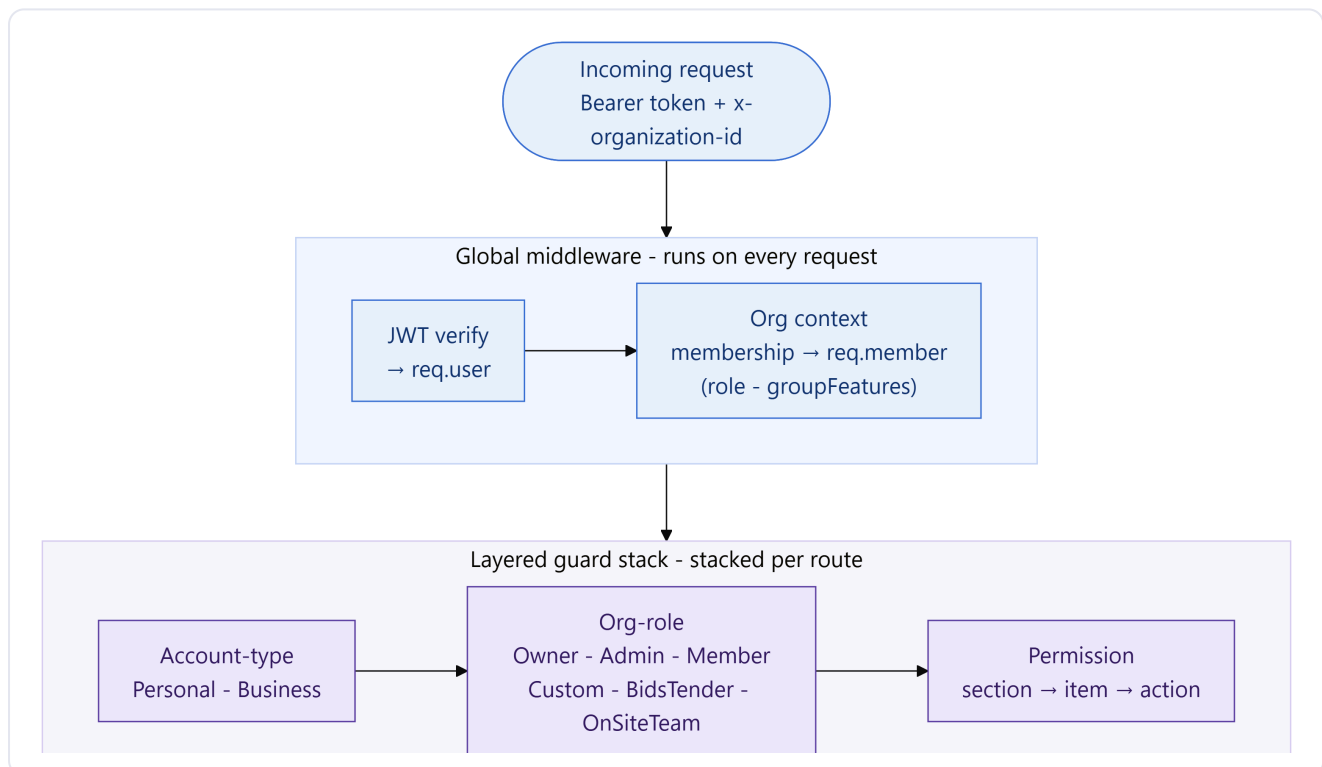
System, transaction & authorization

Original architecture diagrams — no client screenshots or data reproduced.



System architecture — a React SPA over a NestJS modular monolith; every request runs an auth + org-context middleware chain and a stackable guard chain before reaching a domain module.

Bidding & contract lifecycle — pay-to-bid and pay-to-award put real commitment on both sides; completed contracts feed reputation-ranked discovery.



Multi-tenant request & authorization pipeline — tenancy is resolved per request from a header, and small, stackable guards compose the exact access each route requires.

RESULT**Procurement, online end to end****Marketplace**

A directory turned into a monetized, two-sided market

Binding deals

Stripe-verified pay-to-bid & pay-to-award, auto-refunds

Multi-company

One identity, per-org roles & permissions

- **Money built into the funnel.** Pay-to-bid and pay-to-award create real commercial commitment on both sides, with the payment state machine driven by verified Stripe webhooks and idempotent handlers.
- **Multi-tenant from request one.** Header-driven organization context plus a layered guard stack lets one account operate safely across many companies, each with its own role and permission matrix.
- **A trust loop that closes.** Verified reviews roll up into contractor ratings that feed a location-and-reputation ranking powering discovery.
- **Delivery, not just matchmaking.** A shared post-award workspace — tasks, kanban, three finance ledgers, inventory, material requests, messaging, notifications — turns a won bid into a place to run the job.
- **Cloud-native & documented.** An installable, code-split React client and a ~190-endpoint NestJS API deployed on AWS App Runner, with S3-backed media, scheduled jobs, and OpenAPI docs.

Note on scope. Module and endpoint counts reflect the delivered codebase. Client identity, product name, brand assets, proprietary business rules, credentials, and commercial metrics are excluded under NDA; outcomes here are described qualitatively.